Enhancing Password Security via Supervised Learning

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***Abstract*— Text-based passwords are currently the most widely used form of authentication and are expected to remain so. However, since these passwords are typically made up of meaningful strings, machine learning and deep learning algorithms can help developers assess their strengths and predict their vulnerability to brute-force attacks. Advanced techniques such as Long Short-Term Memory (LSTM) and Generative Adversarial Networks (GAN) can generate lists of similar and expected text passwords by learning the patterns of how users create and select their passwords. This paper explores using machine learning algorithms to categorize passwords into three levels: strong, moderate, and weak. We also evaluate whether machine learning or deep learning techniques can learn the patterns used in hashing techniques. Furthermore, we have developed a password generation model using Gated Recurrent Unit (GRU) to create new passwords based on learned patterns. With this approach, we aim to improve password security and enhance the user experience of creating and managing passwords.**

Keywords—password-guessing, GRU, LSTM, GAN, RNN.

I. INTRODUCTION

As a crucial component of various security systems, authentication ways have a wide range of applications. They are extensively used in various industries, such as e-commerce websites, digital computer systems, and new advanced phones. Various methods can be used in authentication, including text-based passwords, tokens, cards, and unique human features like fingerprints and faces. Strings containing alphanumeric characters and additional special characters are all used to build text-based passwords. A token comprises portable USB or smart cards, normally used in multi-factor authentication. Out of all the authentication methods, text passwords are the most commonly used [1] as this method achieved the best availability and can be re-used, plus the very low cost of implementation. With all previous features in text-based, it is still the most important authentication method and will be valid in the future.

While creating strong passwords, security is the goal, and guidelines need to be followed. The National Institute of Standards and Technology (NIST) published updated password rules [2], which include requirements for the minimum number of characters, types of characters, password verification, and limitations on the number of password attempts. Remembering passwords is one of the main challenges for accounts in applications and websites, so users tend to repeat the same passwords for different systems. In recent years, numerous data breaches have resulted in the theft of millions of accounts and passwords. For example, over a billion passwords were stolen in 2019. Some of these breaches are listed in Table 1.

Table 1: A List of famous password breaches globally in 2019. Source: https://haveibeenpwned.com/

Source	*Total of breached passwords*
CafePress	23,205,290
europa.jobs	226,095
Canva	137,272,116
Club Penguin Rewritten	4,007,909
Collection #1	772,904,991
Cracked. to	749,161
EatStreet	6,353,564
EpicBot	816,662

Attacking using the Password guessing method poses a significant security threat to the authentication model based on passwords. Attackers can build a model to learn the pattern of passwords with the help of similar data and repeated passwords. However, the abundance of breached passwords allows attackers to uncover these patterns. These password patterns can serve as guidelines for generating potential passwords in a brute-force attack. For instance, HashCat and John the Rippe are very popular tools for password guessing based on rules [3]. Rule-based password-guessing tools rely on pre-determined patterns or rules to generate candidate passwords, and multiple attempts may be required to guess a password successfully. However, strong and unique passwords and multi-factor authentication can greatly reduce the risk of successful password guessing.

Neural networks are a form of machine learning that can be trained without previous knowledge or assumptions about the data. They have successfully processed speech or natural language and recognized images. Text passwords can be viewed as short phrases and neural networks can learn many features and patterns from leaked passwords. Recent studies indicate that combining neural networks with other password-guessing techniques improves results.

In this paper, we tried many machine learning algorithms to measure the password's strength and suggested a new model to generate potential passwords using deep learning and GRU. The remaining paper is divided as follows—section II lists the most recent related works in password guessing. The methodology to use machine learning in evaluating passwords is explained with results in section III. In section IV, the details of the suggested GRU model are mentioned with the expected results. Finally, we conclude the results and discuss future works in section V.

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Table 2. Summarization of deep learning models used to guess passwords by generating a new set after learning the patterns.

Model Name	***PassGAN***	***SSPG & DPG***	***[12]***	***GENPass***	***TPGXNN***	***[15]***	***BiLSTM RNN***	***PG-RNN***
Technique used	GAN	Transfer Learning & GAN	bidirectional LSTM	PCFG and LSTM	LSTM and VDCNN	LSTM and a semantic analysis	RNN and LSTM	RNN and LSTM

II. BACKGROUND AND RELATED WORKS

Password strength meters can help users generate stronger passwords by guiding password length, complexity, and characters such as digits, symbols, and uppercase letters. Most traditional password strength meters are rule-based, relying on pre-determined guidelines to determine the strength of a password. Deep learning [4] has proven effective in pattern recognition and computer vision tasks. Ian Goodfellow [5] proposed GAN, which consists of a generative and a discriminative model. The generator creates fake samples designed to look like real data, while the discriminator attempts to distinguish between the real and fake samples. The goal of the generator is to produce samples that the discriminator cannot distinguish from real data, and the goal of the discriminator is to identify fake samples accurately. During training, the generator and discriminator are updated based on their performance until the generator produces high-quality, realistic samples. Transfer learning is a technique in deep learning where a pre-trained model is used as a starting point for a new task instead of starting from scratch. This can save time and resources compared to training a model from scratch, as the pre-trained model already has learned features and representations useful for the new task. The pre-trained model is usually fine-tuned for the new task by adjusting some of its layers or adding new layers to handle the specific requirements of the new task.

A recurrent Neural Network is a type of neural network designed to handle sequential data, where the output of a unit depends on its previous state. RNNs have memory-like properties. LSTM [6] is a type of RNN that addresses the vanishing gradient problem, which occurs when training traditional RNNs on long sequences. LSTMs use gates to control the flow of information in and out of the memory cell, allowing them to maintain long-term memory and avoid the vanishing gradient problem. This makes LSTMs well-suited for language modeling, machine translation, and speech recognition tasks.

PassGAN [7] is used to generate password candidates. It works by training a generator network to produce password candidates similar to a dataset of leaked passwords and a discriminator network to distinguish between generated and real passwords. PassGAN is often used in security research to evaluate the strength of passwords and generate new candidates for use in password-cracking attacks. By learning from real-world passwords, PassGAN can generate password candidates more likely to be used by users, making it a useful tool for password cracking and security evaluation.

Dario Pasquini et al. [8] introduced a representation-learning method for password guessing using a GAN; two parts are substring password guessing (SSPG) and dynamic password guessing (DPG). The DPG method dynamically adapts its guessing strategy based on feedback from interacting with targeted password sets. Experiments show that representation learning can effectively apply to password guessing and outperform traditional methods.

Li and others [9] presented a deep neural network incorporating an extended LSTM and a bidirectional language model to model passwords and perform password guessing. The extended LSTM is utilized for modeling the password and extracting features, while the bidirectional language model enhances the performance of the deep neural network. Another example of utilizing deep learning is GENPass [10] which is a deep learning method for password guessing that uses PCFG and LSTM to train and generate candidate passwords. It improves effectiveness through adversarial generation. TPGXNN [11] used the effectiveness of LSTM and VDCNN to train the model with passwords and personal data.

Fang et al. [12] suggest a password-guessing method using a hierarchical semantic model consisting of an LSTM and a semantic analysis model. The semantic analysis component helps the hierarchical semantic model avoid generating non-meaningful substrings, improving performance and efficiency. In [13], Zhang et al. proposes a password-guessing method using structure partitioning and BiLSTM RNN. The structure partitioning module models the passwords in the training set to understand the user's password creation habits and generate a collection of common structures and a sorted string dictionary by probability. This string dictionary serves as the training set for the BiLSTM module.

In addition to LSTM, RNN is also commonly used in password guessing. One is PG-RNN [14], an RNN-based model that automatically dedicates distribution characteristics and character rules from leaked password datasets. We summarized all of the mentioned above models with their deep learning methods or techniques to build the solution in Table 2. Most models utilize LSTM as a base because memory is a property inside LSTM layers.

III. MACHINE LEARNING AND PASSWORDS STRENGTH

Machine learning techniques have been extensively used in the literature to develop several security solutions, such as detecting malicious and phishing websites, defending against dales injection attacks, and identifying darknet tor traffic [15]. A password must be resistant to attacks like brute force attack. A brute-force attack is cracking a password by systematically trying every possible combination of characters until the correct password is found. This can be done manually, by a person trying different combinations, or automatically by using a computer program to try many combinations quickly. A brute-force attack can crack the password for any type of login or encryption, including email, social media, and online banking. In the case of a password, a brute-force attack will typically try all possible combinations of characters for a given length of the password. This means that the longer the password, the more possible combinations there are, and the longer it will take to crack.

However, with the advancement of technology and the availability of powerful hardware and software cracking tools, it has become increasingly easy to perform brute force attacks, and even very strong passwords can be cracked relatively

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quickly. Therefore, it is important to use long and complex passwords and to use different passwords for different accounts. And also use multi-factor authentication, making it much more difficult for an attacker to access an account application. Users can benefit from this technology and the visual variety of products like hairstyles and glasses.

A) A Model to measure Password Meters:

We build a password classification model using Scikit-Learn. The model will give the strength of a password from the lowest strength, labeled 0, to the highest strength, labeled 2. The dataset was collected from different websites through web scraping. The dataset contains around 670000 passwords with different strengths. The dataset was preprocessed by: (1) Checking if any missing values and dropping if so, (2) Shuffle the dataset to understand patterns and relationships within the dataset, and (3) Tokenization: Term Frequency Inverse Document Frequency (TFIDF). This common algorithm transforms the text into a meaningful representation of numbers used to fit machine algorithms for prediction.

Figure 1 displays the main block of our model, and we used two algorithms to train data: Random Forest and Logistic Regression. Logistic regression is a supervised learning classification algorithm used to predict the probability of a target variable as a password in our case. The accuracy achieved is 81%. Random Forest or random decision Forest is an ensemble learning method for classification. The accuracy achieved using Random Forest Classifier is way better than Logistic Regression, which achieved 94.22% accuracy. In addition, we conducted some single predictions for each algorithm and achieved great results in Random Forest with 95% correct predictions out of 200 special tests.

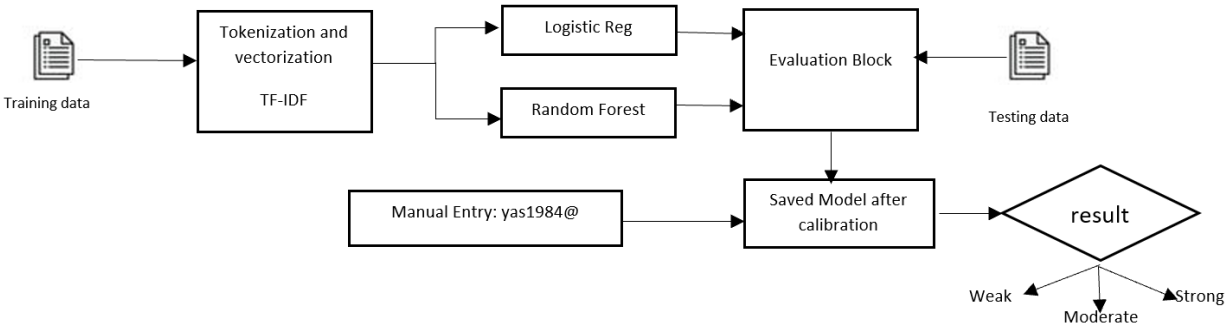

Figure 1: machine learning model for classifying passwords

B) Modul to Learn Hashing type

Password hashing is a process of taking a plaintext password and applying a one-way mathematical function to it to produce a unique fixed-size string of characters called a hash. This hash is then stored in a database rather than the original plaintext password. When a user attempts to log in, the system takes the entered password, applies the same function, and compares the resulting hash to the stored hash. If they match, the login is successful. Password hashing is one-way, meaning it is impossible to reverse the process and obtain the original plaintext password from the hash. This is important because it means that even if someone gains access to the password hash database, they cannot use the hashes to log in to the system or use the plaintext passwords for other purposes.

Different hashing algorithms are available, such as SHA-256, SHA-512, bcrypt, scrypt, argon2, etc. The choice of algorithm depends on the application's specific requirements, such as the level of security needed and the system's performance. A secure and up-to-date hashing algorithm is important to ensure that the hashed passwords are not easily crackable. In our model, we first hash the plain text passwords For Further checking, as Figure 2 shows. We tried to build a model which takes the hashing value as input with a hashing type as a label. The hashing algorithm used is MD5 and SHA1 in the first phase. Then we hashed the same set with SHA2 and SHA512.

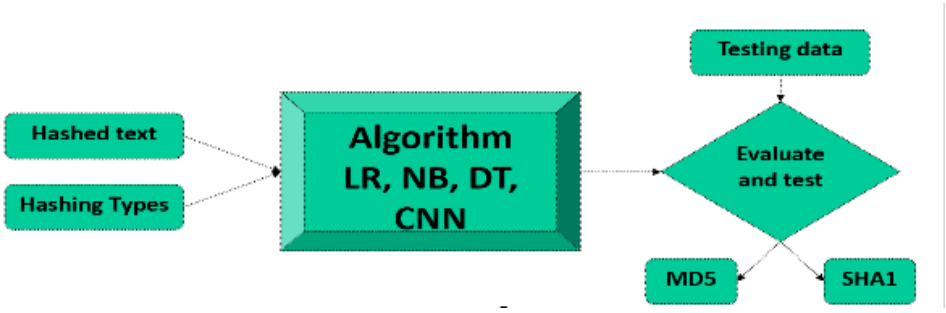

Figure 2: Converting plain text passwords to hashed value

We build the model with machine learning algorithms like Logistic Regression, Naïve Bayes, and Decision Tree Algorithm. Moreover, we tried a deep learning technique by entering data into Convolutional Neural Network (CNN). In the first phase, we tried only two hashing algorithms, and Logistic Regression achieved the best results with 70% accuracy. Others, including CNN, achieve less than 40% accuracy. In the second phase, we increased the number of hashing algorithms to four.

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The results are getting worse as all algorithms are less than 25% accurate. In Table 3, we listed all accuracy results for all algorithms during the first and second phases.

Table 3: Accuracy results while trying to learn hashing type

ML / DL technique	*The first phase (%)*	*Second Phase (%)*
Logistic Regression	70.2	25.1
Naïve Bayes	27	20
Decision Tree Algorithm	40.1	24
CNN	37	24

IV. SUGGESTED GRU MODEL GENERATES PASSWORDS

In our previous research stage, we focused on learning the pattern of password creation. As there is a huge amount of data on leaked passwords, it will help to understand the pattern and how users create their passwords. We used GRU because it uses two gates, the "update gate" and the "reset gate," to control the flow of information.

The update gate decides how much of the previous hidden state should be discarded and how much new information should be added to the hidden state. This allows the network to effectively store and access long-term dependencies in the input sequence. The reset gate determines how much of the previous hidden state should be reset in response to new input. The output of the GRU is then calculated based on the combination of the current input and the previous hidden state and is used as the input for the next step in the sequence. This process allows the GRU to process sequential input data, making it well-suited for language modeling, text generation, and speech recognition tasks.

GRUs are a more computationally efficient alternative to LSTM- RNNs, as they require fewer parameters and reduce the vanishing gradient problem.

V. CONCLUSION AND FUTURE WORKS

Text passwords are combinations of letters and numbers and can be treated as text sequences. Deep learning models have an advantage in processing text data. Recent studies show that deeper learning models are applied in password guessing and password strength analysis. This includes LSTM, RNN, GAN, representation learning, and transfer learning. For password strength analysis, we found that accuracy results are very high by our model and can be enhanced more using different tokenizers. We suggested a new model using a deep learning technique with only GRU for password guessing. The model can identify password patterns and improve guessing performance. This model can generate candidate password sets and assess strength effectively. They have been proven more accurate and practical for text guessing than traditional methods. However, the accuracy and efficiency of deep learning-based methods for password guessing and strength analysis may be limited by the quality of the training datasets. In the future, we will continue implementing the suggested model and tune it with the best hyperparameters.

REFERENCES

[1]. Melicher W, Kurilova D, Segreti SM, Kalvani P, Shay R, Ur B, Bauer L, Christin N, Cranor LF, Mazurek ML. Usability and security of text passwords on mobile devices. In Proceedings of the 2016 CHI Conference on Human Factors in Computing Systems 2016 May 7 (pp. 527-539).

[2]. Lemmon EW, Bell IH, Huber ML, McLinden MO. NIST standard reference database 23: reference fluid thermodynamic and transport properties-REFPROP, Version 10.0, National Institute of Standards and Technology. Standard Reference Data Program, Gaithersburg. 2018.

[3]. Hitaj B, Gasti P, Ateniese G, Perez-Cruz F. Passgan: A deep learning approach for password guessing. InApplied Cryptography and Network Security: 17th International Conference, ACNS 2019, Bogota, Colombia, June 5–7, 2019, Proceedings 17 2019 (pp. 217-237). Springer International Publishing.

[4]. Goodfellow IJ, Pouget-Abadie J, Mirza M, Xu B, Warde-Farley D, Ozair S, Bengio Y. Generative Adversarial Networks, 1–9. arXiv preprint arXiv:1406.2661. 2014.

[5]. Ayub S, Kannan RJ, Alsini R, Hasanin T, Sasidhar C. LSTM-based RNN framework to remove motion artifacts in dynamic multi-contrast MR images with registration model. Wireless Communications and Mobile Computing. 2022 May 4, 2022.

[6]. M. A. Fauzi, B. Yang, and E. Martiri, "PassGAN Based Honeywords System for Machine-Generated Passwords Database," 2020 IEEE 6th Intl Conference on Big Data Security on Cloud (BigDataSecurity), IEEE Intl Conference on High Performance and Smart Computing, (HPSC) and IEEE Intl Conference on Intelligent Data and Security (IDS), Baltimore, MD, USA, 2020, pp. 214-220, doi 10.1109/BigDataSecurity-HPSC-IDS49724.2020.00046.

[7]. Pasquini D, Gangwal A, Ateniese G, Bernaschi M, Conti M. Improving password guessing via representation learning. In2021 IEEE Symposium on Security and Privacy (SP) 2021 May 24 (pp. 1382-1399). IEEE.

[8]. Li T, Jiang Y, Lin C, Obaidat MS, Shen Y, Ma J. Deepag: Attack graph construction and threats prediction with bi-directional deep learning. IEEE Transactions on Dependable and Secure Computing. 2022 Jan 18;20(1):740-57.

[9]. Xia, Zhiyang, Ping Yi, Yunyu Liu, Bo Jiang, Wei Wang, and Ting Zhu. "GENPass: a multi-source deep learning model for password guessing." IEEE Transactions on Multimedia 22, no. 5 (2019): 1323-1332.

[10]. Zhou H, Liu Q, Zhang F. Poster: An analysis of targeted password guessing using neural networks. In Proceedings of the 2017 IEEE Symposium on Security and Privacy (S&P) 2017.

[11]. Fang Y, Liu K, Jing F, Zuo Z. Password guessing based on semantic analysis and neural networks. InTrusted Computing and Information Security: 12th Chinese Conference, CTCIS 2018, Wuhan, China, October 18, 2018, Revised Selected Papers 12 2019 (pp. 84-98). Springer Singapore.

[12]. Zhang, Yi, Hequn Xian, and Aimin Yu. "CSNN: Password guessing method based on Chinese syllables and neural network." Peer-to-Peer Networking and Applications 13 (2020): 2237-2250.

[13]. Bai Q, Zhou J, He L. PG-RNN: using position-gated recurrent neural networks for aspect-based sentiment classification. The Journal of Supercomputing. 2022 Feb;78(3):4073-94.

[14]. Abu Al-Haija, Q., Al-Fayoumi, M. An intelligent identification and classification system for malicious uniform resource locators (URLs). Neural Comput & Applic (2023). https://doi.org/10.1007/s00521-023-08592-z

[15]. Bai Q, Zhou J, He L. PG-RNN: using position-gated recurrent neural networks for aspect-based sentiment classification. The Journal of Supercomputing. 2022 Feb;78(3):4073-94.

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